

Microprocessor Based System Design Laboratory (Gx)

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Experiment No. 3: 8085 Assembly Language Programming (Continued)

Objective: To write simple programs in 8085 Assembly Language and test them at the 8085 SDK (System Development Kit).

Preamble: By this time you are familiar with

- (a) writing simple 8085 assembly language programs,
- (b) translating them into 8085 machine language,
- (c) loading them in the RAM section of the SDK,
- (d) executing them, and finally
- (e) testing their effects to validate their “correctness”.

Following are a few examples where **simple C program segments** are “implemented” by 8085 programs. Please consult the **2-page Instruction Set** and **16-page Instruction Set Reference Encyclopedia** for validating the programs; and read the comment(s) against each instruction of the programs.

Please try to execute and validate the following programs.

Example 1. C statement: `x = 5;` /* *x is an integer variable* */

Let the memory location **8800H** be used for the variable **x**.

Assembly Instruction	Memory Address (Hex)	Machine Language (Hex)	Comment
Alternative 1			
Data Area {	8800	--	Memory location for variable x . Note that x is not initialized.
MVI A,05H	8801	3E	Puts 5 in A.
	8802	05	
STA 8800H	8803	32	Stores contents of A in memory location 8800H
	8804	00	
	8805	88	
HLT or RST 3	8806	76 or DF	HLT halts 8085. You have to “ RESET ” the kit to make the kit working again. Check the contents of the memory location 8800H (by “ EXAM MEM ” key press), which should be 5. If you use RST 3 (in place of HLT), the Interrupt Service Routine (ISR , at address 0018H) in your kit for the software interrupt instruction RST 3 saves the contents of all the registers of the 8085 microprocessor at fixed memory locations. Subsequently when you press “EXAM REG” key, the monitor program shows the-then contents of the registers, taking them from those locations of the memory. Now you can check the contents of the register A (by “ EXAM REG ” key press) and the memory location 9000H (by “ EXAM MEM ” key press), both should be 5. Check contents of PC too.
Alternative 2			

Data Area {	8800	--	Memory location for variable x . Note that x is not initialized.
LXI H, 8800H	8801	21	Puts 8800H in HL register pair.
	8802	00	
	8803	88	
MVI M,05H	8804	36	
	8805	05	
HLT or RST 5	8806	76 or EF	See comments for this in Alternative 1 . Check contents of 8800H location. or See comments for this in Alternative 1 . Check contents of A, PC and 8800H .

Example 2: C Program Segment: -

```
int x = 0;
```

```
int y;
```

```
y = (x > 5); /* "x > 5" being false, y should contain false (ie, 0) */
```

Let the memory locations 8800H and 8801H be used for the variables **x**, and **y**, respectively.

Label	Assembly Instruction	Memory Address (Hex)	Machine Language (Hex)	Comment
Data Area {		8800H	00	Memory location for x which is initialized to 0.
		8801H	--	Memory location for y which is not initialized.
⋮				
code may start here				
	LDA 8800H	8900	3A	Loads contents of memory location 8800H (variable x) in register A.
		8901	00	
		8902	88	
	CPI 05	8903	FE	Compares A and 05H.
		8904	05	This instruction sets Z flag (to 1) if (A) == 5 (ie, if contents of A is 5) CY flag (to 1) if (A) < 5, (ie, if contents of A is less than 5)
	LXI H, 8801H	8905	21	HL register pair now stores address of y
		8906	01	
		8907	88	
	JC Zero	8908	DA	Jump to the label ZERO if CY is 1, that is, if A < 5
		8909	13	
		890A	89	
	JZ Zero	890B	CA	Jump to the label ZERO if Z is 1, that is, if A = 5
		890C	13	
		890D	89	
	MVI M,01H	890E	36	Stores 1 in y
		890F	01	
	JMP End	8910	C3	
		8911	15	
		8912	89	
Zero	MVI M,00H	8913	36	Stores 0 in y
		8914	00	
End	HLT	8915	76	See comments for this in Example 1 . Check contents of 8801H location.

	or		or	or
	RST 5		EF	See comments for this in Example 1 . Check contents of A, PC and 8801H.

Try your program with different values of x (say, x = 2, 9) by changing the contents of 8800H and then executing the program.

Example 3: C program segment -

```

int x;
int y;
x = 9;
y = 5;
if (x == y)
    z = 5;
else
    z=10;

```

where, x, y and z are integer variables.

In the 8085 program (given below) variables x, y and z are “implemented” as registers B, C, and D, respectively.

Note that there is no 8085 instruction for comparing registers B and C. Comparison is always with register A. Contents of B can be copied to A, so that, A and C can be compared later to do the job.

Label	Assembly Instruction	Memory Address (Hex)	Machine Language (Hex)	Comment
Start:	MVI B, 09H	8800	06	For x = 9; statement
		8801	09	
	MVI C, 05H	8802	0E	For y = 5; statement
		8803	05	
	MOV A, B	8804	78	Contents of B is copied to A.
	CMP C	8805	B9	Compares A and C. This instruction sets Z flag (to 1) if (A) == (C), ie, contents of A and C are same sets CY flag (to 1) if (A) < (C), ie, contents of A is less than C
	JZ MakeD5	8806	CA	If Z flag is 1, ie, in this case, (B)==(C) then jump to label MakeD5 (Address 880EH), where D will be made 5, ie, D=5
		8807	0E	
		8808	88	
MakeD10:	MVI D,0AH	8809	16	The else-part of the C program segment. Make D=10.
		880A	0A	
	JMP End	880B	C3	Following instruction implements the if-part of the C program segment. Hence, jump to End (Address 8810H) to skip that part.
		880C	10	
		880D	88	
MakeD5:	MVI D,05H	880E	16	The if-part of the C program segment. Make D=5.
		880F	05	
End:	RST 5	8810	EF	See the comments for this in Example 1 above. Note HLT instruction here (in place of RST 5 will not let you check the-then values of x, y, and z (ie, registers B, C, and D)

After execution of the above program (GO-8-8-0-0-EXEC) you may check the-then contents of registers A, B, C, D, PC, etc to validate the program.

Change and try the above program for different values of x and y (that is, registers B and C)

- **Please follow the tabular format of the above examples while writing your programs for the problems given below.**
- **You may, however, defer writing comments until you have a correct running program.**

In today's laboratory class **each of you** have to

1. **try the example programs given above**
2. write 8085 programs as specified below **in the format shown in the examples**. For each program, you **(i)** write the assembly code first, **(ii)** and then translate it to machine language using the Instruction Set Table provided to you, **(iii)** and finally you load, execute, and test the program.
3. **Get the program, that you have written in your notebook, signed by your teacher. Take a photo of your program and submit it at your Google Classroom.** The problems statements are available as classwork in the Classroom.

Programming Assignments

[Consult the **2-page Instruction Set** and **16-page Instruction Set Reference Encyclopedia** for choosing appropriate Instructions while writing programs]

1. To do what the following C program segment does. [Similar to Example 3. above]

```
int x = 5;
int y = 15;
int z;
z = y + (2*x > y)? 10: 5;
```

where, x, y and z are integer variables. Use memory locations to implement these variables (say, **8800H**, **8801H**, **8802H** for x, y, and z, respectively.) Please note that, as per C language, **(2*x >= y)? 10: 5** is a **conditional expression**, and its value is **10** if the condition **(2*x >= y)** is true; otherwise its value is **5**.

Modify your program for different initial values for x and y and check the results.

2. To do what the following C program segment does. [Similar to Example 2. above]

```
int x = 9;
int y = 7;
int z;
z = (x & y); /* bitwise anding */
```

Let the memory locations **8800H**, **8801H**, and **8802H** be used for the variables x, y and z, respectively.

Change and run your program for different initial values of x and y.

3. Repeat assignment 1. above with relational expression "**2*x < y**" (in place of "**2*x > y**").
4. Repeat assignment 1. above with relational expression "**2*x == y**" (in place of "**2*x > y**").
5. Repeat assignment 1. above with relational expression "**2*x != y**" (in place of "**2*x > y**").
6. Repeat assignment 1. above with relational expression "**2*x >= y**" (in place of "**2*x > y**").
7. Repeat assignment 1. above with relational expression "**2*x <= y**" (in place of "**2*x > y**").

8. To do what the following C program segment does.

```
char ch='C', dept[10];
switch (ch) {
    case 'A': strcpy(dept, "ARCH");
                break;
    case 'C': strcpy(dept, "CST");
                break;
    default: strcpy(dept, "IESTS");
}
```

where, **ch** is a character variable mapped to memory location **8800H**, and **dept[10]** is a character array mapped to memory locations **8801H to 880AH** in the memory. **Please note that characters are stored as their respective ASCII values.** That is 'A' is stored as 65 (41H) and "CST" needs 4 Bytes to store 67, 83, 84, 0 (that is, 43H, 53H, 54H, 00H). **Change and run your program for different initial values of ch.**

Writing Subroutines (Functions) in 8085

Example 4: C Function *int larger(int x, int y)* returns whichever is larger between x and y.

```
int larger(int x, int y) {
    if (x > y)
        return (x);
    return (y);
}
```

A functionally similar 8085 assembly language function has to be written. Let parameters (x and y) be passed to the function through memory locations **8800H** and **8801H**, respectively; and the larger value is returned through the register **B**.

Label	Assembly Instruction	Memory Address (Hex)	Machine Language (Hex)	Comment
Parameters		8800H	--	Memory location for the parameter x
		8801H	--	Memory location for the parameter y
⋮ code may start here				
larger:	LXI H, 8800H		21	Load HL pair with address of the parameter x (Address 8800H)
			00	
			88	
	MOV A,M		7E	Copy x to A
	INX H		23	Make HL pair store the address of y (Address 8801H)
	MOV B,M		48	Copy y to B
	CMP B		B8	Compare B (ie, y) with A (x) This instruction sets Z flag (to 1) if (A) == (B), ie, if x = y CY flag (to 1) if (A) < (B), ie, if x < y
	RC		D8	Return on Carry flag , B (ie, y) already contains the larger number
	MOV B,A		47	Make B store the larger (A , ie, x)
	RET		C9	

main:	LXI H, 8800H		21	Load HL pair with address of the parameter x (Address 8800H)
			00	
			88	
	MVI M, 15H		36	Keep the actual parameter 15H for x
	INX H		23	Make HL pair store the address of y (Address 8801H)
	MVI M, E3H		36	Keep the actual parameter E3H for y
	CALL larger		CD	Call the “ larger ” function (available at 8800H)
			00	
			88	
	MOV A,B		78	Copy the return value to the register A.
	STA 8802H		32	Save the return value at location 8802H.
			02	
			88	
	HLT or RST 5		76 or EF	Check contents of 8802H to see the result.

For each of the following functions, you have write a caller program/function also.

9. Write an 8085 subroutine (function) which is functionally same as “*int isupper(char c)*” as defined below.

```
int isupper(char c) {
    if (c >= 'A' && c <= 'Z') return(1);
    return(0);
}
```

10. Write an 8085 subroutine (function) which is functionally same as “*int islower(char c)*” as defined below.

```
int islower(char c) {
    if (c >= 'a' && c <= 'z') return(1);
    return(0);
}
```

11. Write an 8085 subroutine (function) which is functionally same as “*int isalpha(char c)*” as defined below.

```
int isalpha(char c) {
    return( isupper(c) || islower(c)); /* it calls previously defined isupper() and islower()
functions */
}
```